

Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2015–2025)

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Contents

1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people worldwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinson et al., 2020 found that 69% of Deaths due to a Generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity render it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.

However, translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the intricate and non-linear complex temporal dynamic of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations.

This review therefore synthesizes and critically evaluates recent advances in time-series deep learning—particularly convolutional neural networks (CNNs) and long short term memory (LSTMs)—applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and rigorous, prospective evaluation can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

subsectionResearch Aim and Questions How do different deep learning architectures and biosignal modalities excluding EEG compare in their ability to achieve a optimal trade-off between sensitivity and false alarm rate for ambulatory seizure monitoring?

2 Theoretical Foundations for Ambulatory Seizure Monitoring

2.1 The Ambulatory Monitoring Paradigm: Requirements and Challenges

beniczkyAutomatedSeizureDetection2021a, chenSeizuresDetectionUsing2022a found that translation of seizure detection from the controlled clinical environment to an ambulatory setting redefines approaches and performance requirements. In clinical environments maximum sensitivity is prioritized over FAR, when engineering for ambulatory use a balance of both metrics is needed. Failing to detect a seizure (low sensitivity) poses a safety risk, While excessive false alarms (high FAR) leads to desensitizing and "alarm fatigue" and the device being ignored or not used at all. Users reported that devices should achieve high sensitivity ($\geq 90\%$) with low false alarm rates — acceptable FARs were 1-2/week for patients with ≥ 1 seizure/week and 1-2/month for patients with < 1 seizure/week (Sivathamboo et al., 2022). While more complex models can find more intricate patterns, they also introduce the "black-box" problem wherein the model lacks transparency AbuAlrob et al., 2025; Ghaderi, 2025. Therefore, the design should incorporate explanations for decisions to ensure user trust and regulatory compliance AbuAlrob et al., 2025; Miron et al., 2025.

2.2 Physiological Biosignals for Wearable Sensing

Wearable devices circumvent the impracticality of EEG by capturing peripheral biosignals that serve as proxies for central nervous system activity during seizures. These proxies will inherently relay incomplete and noisy signals so a combination of multiple sensors is required. The acceleration/attitude sensor excels at detecting muscle convulsions but is noisy in daily use, as it captures all movement. The electrodermal activity (EDA) sensor detects autonomic arousal by measuring the skin conductance associated with sweat gland activity.

2.3 Seizure Manifestations and Corresponding Biosignals

Wearable biosignals act as peripheral proxies for cerebral activity during seizures. Different seizure types produce distinct physiological signatures across these signals, informing detection approaches. This relationship centers on two primary manifestations: motor activity and autonomic activation.

Convulsive seizures, such as generalized tonic-clonic events, produce clear motor signatures captured by accelerometry (ACC), which measures movement and posture changes characteristic of ictal events (Wang et al., 2025; Wu et al., 2024).

Simultaneously, seizures drive autonomic responses recorded through complementary biosignals. Electrodermal activity (EDA) measures sympathetic nervous system arousal via skin conductance changes (Jain et al., 2017; Miron et al., 2025). Cardiac activity provides particularly strong autonomic proxies: electrocardiography (ECG) captures heart rate and rhythm alterations (Ghaderi, 2025; Leal et al., 2017), while heart rate variability (HRV) quantifies autonomic balance dynamics (Mason et al., 2024; Pavei et al., 2017).

In wearable applications, photoplethysmography (PPG) serves as a practical alternative for cardiac monitoring, deriving pulse rate and variability though with greater motion

sensitivity than ECG ([chenSeizuresDetectionUsing2022a](#); Seth et al., [2023](#)).

Consequently, detection strategies combine these biosignals according to target seizure types, with multimodal fusion providing the most robust approach for ambulatory monitoring.

2.4 Evaluation Frameworks: From Retrospective Analysis to Prospective Validity

The clinical validity of a seizure detection algorithm is determined by the rigor of its evaluation, formalized in the ILAE’s phased framework (Phases 0–4) ([beniczkyAutomatedSeizureDetecti](#)). Retrospective studies (Phases 0-2), while foundational, risk significant data leakage and optimistic bias if they use non-chronological data splits, allowing future information to inform training (Kalousios et al., [2024](#); Wong et al., [2023](#)). Pseudo-prospective validation addresses this by enforcing chronological order, training only on past seizures to test on subsequent ones, providing a more realistic performance estimate (Kalousios et al., [2024](#); Lu et al., [2025](#)). The clinical gold standard is a prospective trial (Phase 3), which requires a locked algorithm, a video-EEG reference standard, and testing on new, unseen data from multiple centers. The final step is in-field evaluation (Phase 4) in patients’ homes, where metrics like the false alarm rate per hour (FAR/h) become decisive for practical usability, as demonstrated by long-term studies like Nightwatch (Miron et al., [2025](#); Shum & Friedman, [2021](#)).

3 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, [2002](#)) and PRISMA principles (Tugwell & Tovey, [2021](#))). The methodological approach uses a multi-stage search and transparent extraction/synthesis procedures as outlined below.

1. Search strategy (Multi-stage, 2015–2025):

- *Scoping*: Google Scholar for broad coverage and identifying candidate papers.
- *Formal queries*: IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
- *Citation chasing*: Backward and forward citation tracking on key papers to ensure completeness.

2. Inclusion / Exclusion criteria:

- *Inclusion*: Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
- *Exclusion*: EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate a concept matrix with the following fields:

1. Setting: clinical vs ambulatory.
2. Modality: ECG/HRV, PPG, EDA, ACC.

3. Task: detection vs prediction.
4. Model class and fusion strategy.
5. Study phase: retrospective, pseudo-prospective, prospective.
6. Evaluation granularity: sample- vs alarm-based.
7. Windowing specifics (window length, stride, overlap).
8. Metrics: sensitivity, FAR/h, AUC, f1-score, latency.
9. Personalization.
10. Interpretability and safety considerations.
11. Data characteristics and dataset identity.
12. Missing data handling (imputation, exclusion) and synthetic data use.
13. Inference requirements (latency, computational load, hardware).

Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

4 Search Strategy

4.1 PICO-Based Search Methodology

The literature search was structured using the PICO framework (Patient/Problem, Intervention, Comparison, Outcome) to identify relevant studies on deep learning for non-EEG epilepsy detection. The following key terms were used in combination with Boolean operators (AND, OR, NOT):

1. **P (Patient/Problem):** Epileptic seizures in need of detection or prediction.
 - *Keywords:* ‘epilepsy’, ‘seizure’, ‘ictal’, ‘preictal’
 - *MeSH Terms (PubMed):* Epilepsy, Seizures
2. **I (Intervention):** The diagnostic method based on wearable biosignals and deep learning.
 - *Biosignal Keywords:* ‘electrocardiogram’, ‘ECG’, ‘heart rate variability’, ‘HRV’, ‘photoplethysmography’, ‘PPG’, ‘electrodermal activity’, ‘EDA’, ‘accelerometer’, ‘ACC’
 - *Technology Keywords:* ‘wearable’, ‘wearable device’
 - *Method Keywords:* ‘deep learning’, ‘machine learning’, ‘convolutional neural network’, ‘CNN’, ‘long short-term memory’, ‘LSTM’, ‘neural network’
 - *MeSH Terms (PubMed):* Wearable Electronic Devices, Electrocardiography, Deep Learning

3. **C (Comparison):** The explicit exclusion of studies focused solely on electroencephalography (EEG) as the primary modality.
 - *Exclusion Keywords:* ‘EEG’, ‘electroencephalography’, ‘iEEG’, ‘intracranial’
 - *Search Logic:* The NOT operator was applied to this group in all database searches.
4. **O (Outcome):** Metrics quantifying the performance of a detection or prediction system.
 - *Keywords:* ‘detection’, ‘prediction’, ‘sensitivity’, ‘specificity’, ‘false alarm rate’, ‘FAR’, ‘area under the curve’, ‘AUC’

/subsectionExploration

Due to the insufficient amount of papers gathered from systematic review an exploration was done on the already existing papers. The exploration consisted of graph tools to visualize backlinks and gemini deep research feature to find relevant papers. Through this method an additional 4 papers were acquired bringing the total number of papers up to 13.

4.2 Practical Filters and Reproducibility

Consistent with the overall review scope, searches were restricted to **2015–2025** and **journal articles** with few exceptions made regarding **conference proceedings**. All database searches were documented (query string, platform, and execution date). Results were deduplicated prior to screening.

Note on exclusions: Because some non-EEG wearable studies rely on video-EEG for labeling, EEG-related terms were used to filter *EEG-only modalities*, not to exclude studies that mention EEG solely as a reference standard.

5 Results

5.1 Study Selection and Characteristics

Following the PRISMA guidelines, 13 studies published between 2013 and 2026 were included in this systematic review. All studies focused on non-EEG wearable devices for epileptic seizure detection or forecasting. Table ?? provides a comprehensive comparison of study characteristics following the Webster and Watson **webster2002** framework.

5.2 Comparison of Studies

5.3 Detection Performance Summary

5.3.1 Sensitivity Analysis

Detection studies reported sensitivities ranging from 71.6% (**dong2026**) to 100% (**fine2025**; **borujeny2013**). The highest sensitivity (100%) was achieved by **fine2025**<empty citation> for tonic seizure detection (Phase 1, n=3 test patients) and **borujeny2013**<empty citation> using KNN (k=5) on a very small sample (n=3, 20 seizures). However, these results must be interpreted with caution given the limited sample sizes.

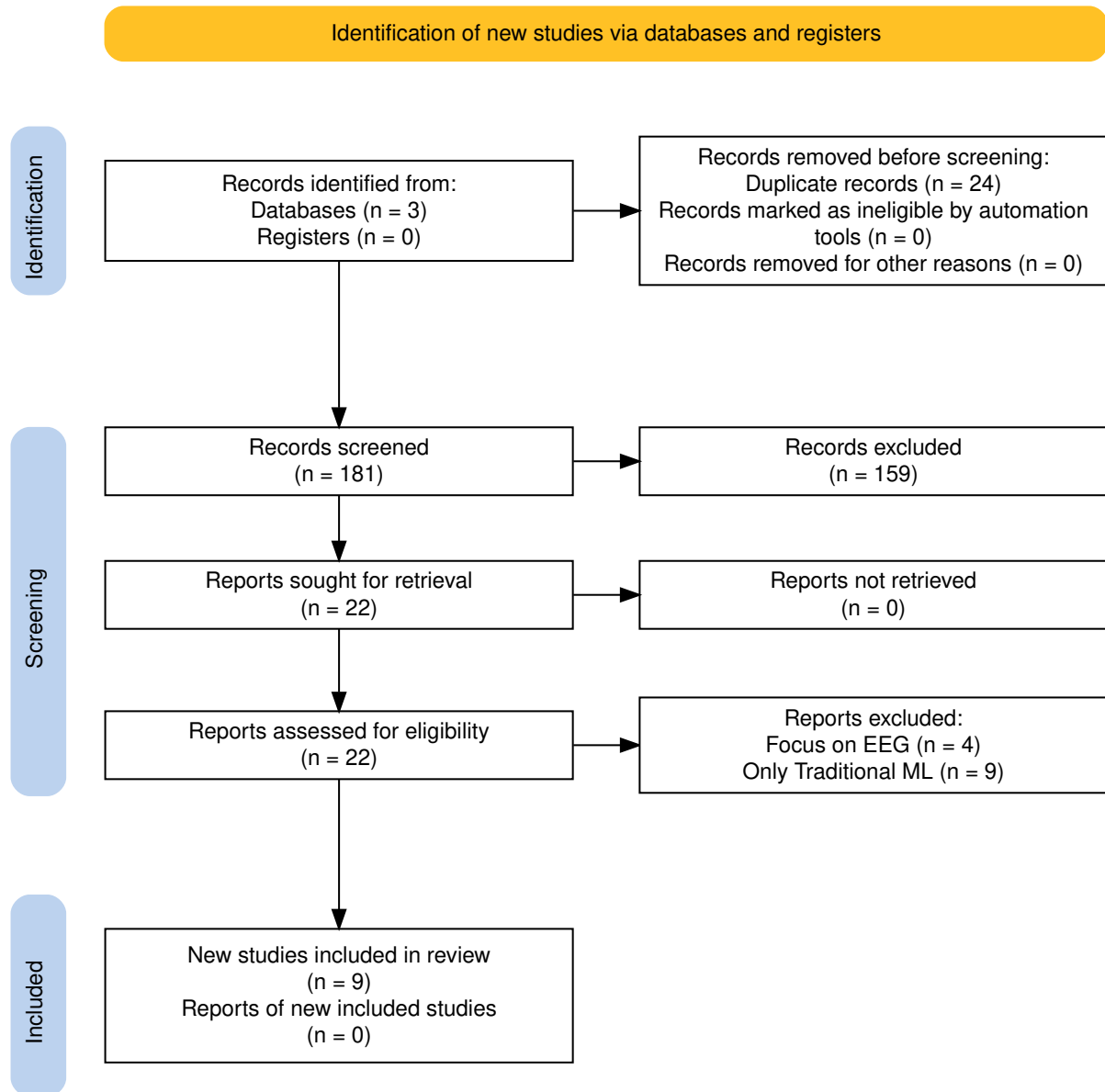


Figure 1: PRISMA flow diagram of the study selection process.

Table 1: Comparison of 13 studies on wearable seizure detection and forecasting

Study	Objective	Design	Sample	Seizure Type	Device	Location	Algorithm	Sens
<i>Detection Studies</i>								
Reintjes 2025	ECG-based anomaly detection	Retrospective train/test	n=120 856 sz	Focal	ECG (Sensor Dot)	Chest	Matrix Profile TimeVQV(SDW)	98.2%
Fine 2025	Tonic sz detection	Phase 1	n=15/3 train/test	Tonic motor	Wristband (6-axis)	Wrist	ANN (594 feat)	100% (test)
Spahr 2025	Convulsive sz detection	Prospective multi-center	n=384 49 CSs	Gen/ bilateral	Empatica (ACC)	Wrist	Ensemble 1D CNN	96% (90–100)
Dong 2026	Nocturnal severe sz	Prospective 10-fold CV	n=68 788 nights 1846 sz	TC, hypermotor tonic>30s	NightWatch armband	Upper arm	Two-stage CNN-LSTM	71.6% overall 33.4%)
Wang 2025	Attitude angle	Retrospective 10-fold CV	n=28 62 sz	Motor	Biovital-P1 (EMP+watch)	Wrist	LSTM	Acc: 97.8%
Singh Rathore 2024	Multimodal detection	Retro single-pt	1 pt 6 hrs	–	EDA+ACC+HR+BVP	–	MLP, SVM KNN, NB, LR	96.8% (MLP)
Borujeny 2013	Wireless accel	Retro 3-patient	3 pts 20 sz	Motor	3times2D ACC	Arms, thigh	ANN, KNN	100% (KNN k=5)
Elemam 2025	Q+HRV+audio	Cross-sectional	Q:198 HRV:30	Gen+PNES	Smartphone camera	Finger/thumb	CNN, rule-based	95.1% (Q)
<i>Forecasting Studies</i>								
Nasserri 2021	Ambulatory forecasting	Retro of prospective	n=6	Focal, FBTC	Empatica E4	Wrist	LSTM RNN (4 layers)	AUC 0.75 (0.50–0.92)
Stirling 2021	HR cycle forecaster	Retro+pseudo	n=11 136 sz	Focal (8) Gen (1) Both (2)	Fitbit	Wrist	LSTM+RF+LR ensemble	
Meisel 2020	No pt-specific training	Retro, LOSO CV	n=69 452 sz	Focal, Gen	Empatica E4	Wrist/Ankle	LSTM	51.2% mean 75.6%*
Ode 2023	RRI prediction	Retro validation	n=66 264 inter 85 pre	Focal (TLE, ELF)	ECG (RRI)	–	SA-AE	74% 100% in 29/66

reintjes2025<empty citation> achieved 98.2% sensitivity for ECG-based detection using Matrix Profile, but with a false alarm rate of 13.9 FA/h—far above clinically acceptable levels. **spahr2025**<empty citation> reported 96% sensitivity (95% CI: 90–100%) for convulsive seizure detection using a single accelerometer sensor, with a favorable false alarm rate of 1/61 nights.

5.3.2 Device Types and Locations

Wrist-worn devices were the most common (7/13 studies), utilizing various sensor combinations:

- Accelerometer-only: **spahr2025**<empty citation>
- Multi-modal (ACC + EDA + BVP + TEMP): **meisel2020**; **nasseri2021**<empty citation>
- ECG/single-lead: **reintjes2025**; **ode2023**<empty citation>

5.4 Forecasting Performance Summary

Four studies focused on seizure forecasting rather than detection. **nasseri2021**<empty citation> achieved the highest forecasting performance (AUC 0.75, range 0.50–0.92) using LSTM RNN with long-term (median 220 days) ambulatory monitoring. **stirling2021**<empty citation> reported AUC 0.74 for hourly forecasts, with 100% of patients performing above chance.

meisel2020<empty citation> demonstrated that 43.5% of patients (30/69) achieved significantly better-than-chance forecasting without patient-specific training, with mean sensitivity of 75.6% in this subgroup. **ode2023**<empty citation> achieved 74% sensitivity using self-attentive autoencoder on R-R intervals, with 100% sensitivity in 29/66 patients.

5.5 Key Limitations Across Studies

Several recurring limitations were identified:

1. **Small sample sizes:** 6/13 studies had $n < 20$; only 4 studies had $n > 100$
2. **Controlled settings:** Most detection studies were conducted in EMU environments, limiting generalizability to real-world settings
3. **Retrospective analysis:** 8/13 studies used retrospective designs, even when data was collected prospectively
4. **Seizure type specificity:** Many studies focused only on convulsive or motor seizures, limiting applicability to non-motor seizure types
5. **Offline processing:** 5/13 studies explicitly mentioned offline analysis, not real-time implementation

5.6 Emerging Trends (2013–2026)

Analysis of publication trends reveals:

- Shift from traditional ML (ANN, KNN in 2013) to deep learning (LSTM, CNN dominant post-2020)
- Increased focus on ambulatory/home monitoring (post-2020)
- Multi-modal sensor fusion becoming standard
- Tunable sensitivity algorithms emerging for clinical customization (**spahr2025**)
- Diary + wearable hybrid approaches for daily-resolution monitoring (**vieluf2025**)

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